19 $2 - \dfrac{x+2}{x-3} - \dfrac{x-6}{x+3}$ can be written as a single fraction in the form $\dfrac{ax+b}{x^2-9}$ where a and b are integers.

Work out the value of a and the value of b.

$$\frac{2(x-3)(x+3) - (x+2)(x+3) - (x-6)(x-3)}{(x-3)(x+3)}$$

1 mark
Write all three terms over the common denominator

×	x	$+3$
x	x^2	$+3x$
-3	$-3x$	-9

×	x	$+3$
x	x^2	$+3x$
$+2$	$+2x$	$+6$

×	x	-3
x	x^2	$-3x$
-6	$-6x$	$+18$

×	x^2	-9
$+2$	$+2x^2$	-18

Denominator: $x^2 - 9$

Numerator: $2x^2 - 18 - (x^2 + 5x + 6)$

$- (x^2 - 9x + 18)$

1 mark
Correct bracket expansions

Numerator = $2x^2 - 18 - x^2 - 5x - 6 - x^2 + 9x - 18$

$= 4x - 42$

1 mark
Subtract every term in each bracket correctly

1 mark
a = 4 and b = −42

$a =$ 4

$b =$ −42

(Total for Question 19 is 4 marks)

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